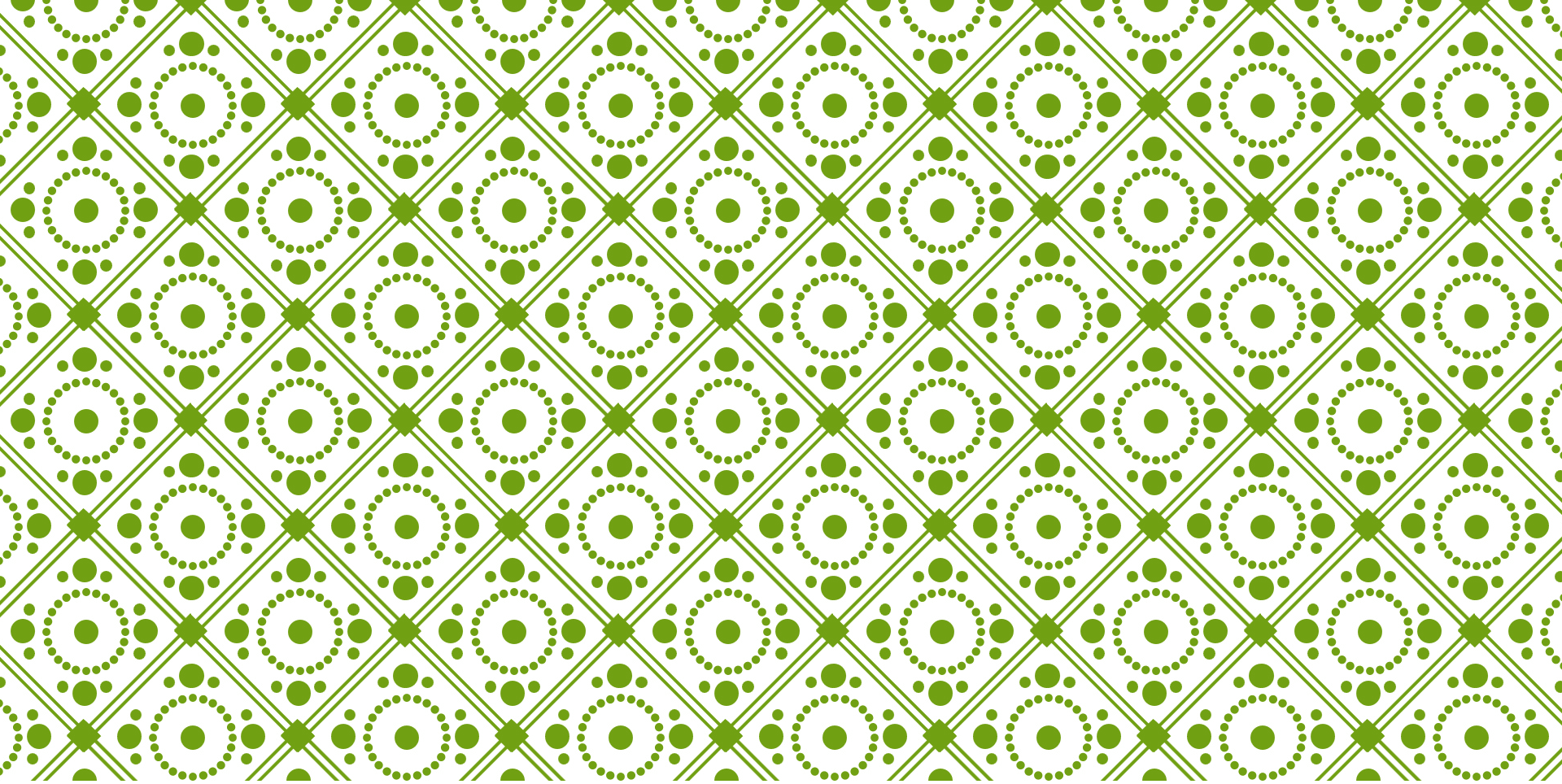




DERIVATIVES

DEFINITION

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2021 - 2022

Definition

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Note:

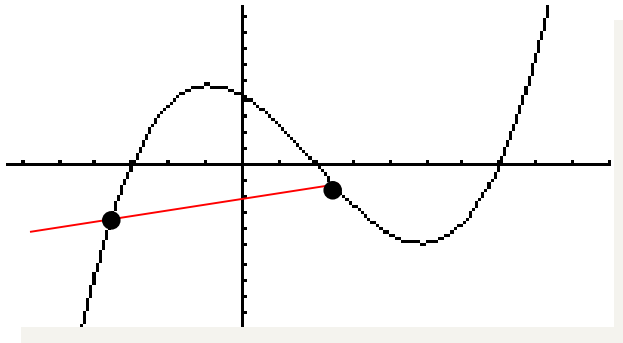
- The derivative is the rate of change function for $f(x)$
- The derivative is also a function of x
- The limit must exist
- Other notations besides $f'(x)$ include: $\frac{dy}{dx}$, y' , $\frac{d}{dx}[f(x)]$, $D_x[y]$

Average and Instantaneous Velocity

Difference Quotient

- $$\frac{f(a) - f(b)}{a - b}$$

- Slope of secant

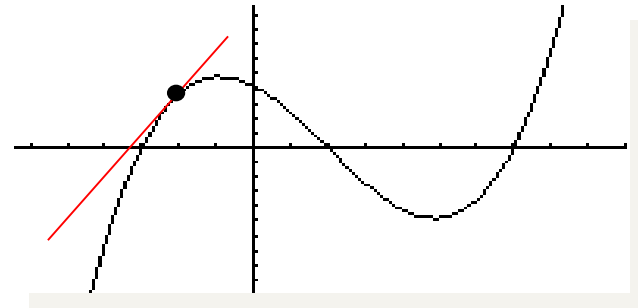


- Average rate of change
- Average velocity

Derivative

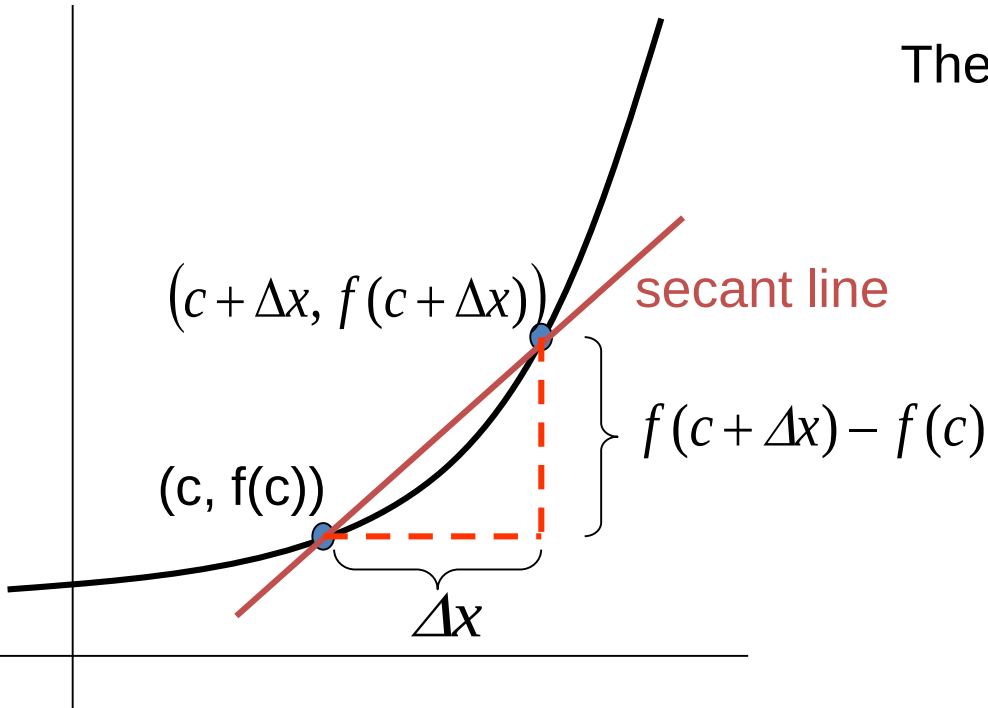
- $$\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

- Slope of tangent



- Instantaneous rate of change
- Instantaneous velocity

The Tangent Line Problem

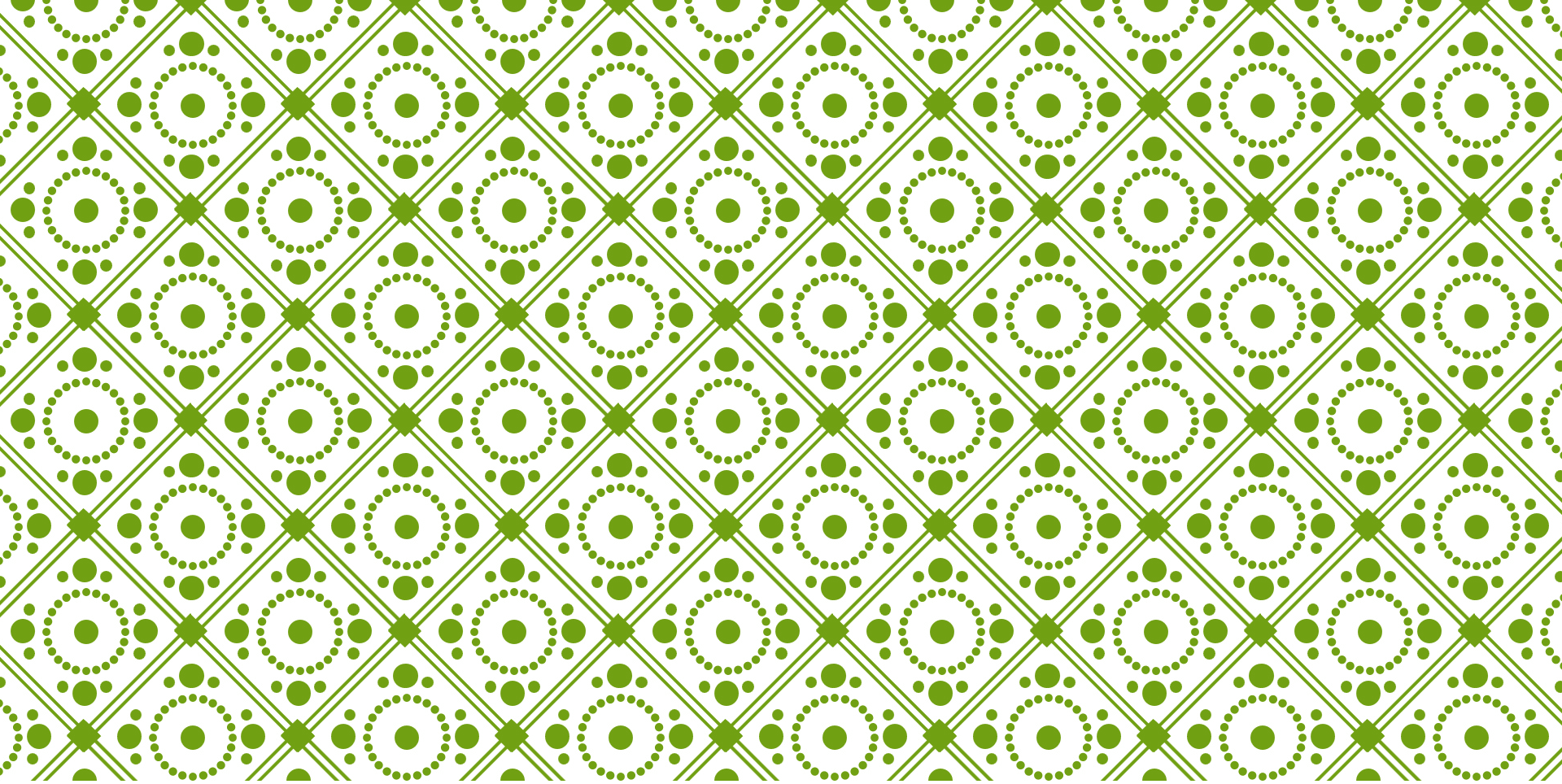


The slope between these two points is

$$\begin{aligned} m_{\text{sec}} &= \frac{f(c + \Delta x) - f(c)}{c + \Delta x - c} \\ &= \frac{f(c + \Delta x) - f(c)}{\Delta x} \end{aligned}$$

Tangent Line with Slope m :

$$\lim_{\Delta x \rightarrow 0} \frac{f(c + \Delta x) - f(c)}{\Delta x} = m$$



DERIVATIVES

DERIVATIVES OF FUNCTIONS

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2021 - 2022

Derivatives of Functions

$$1. \frac{d}{dx}[cu] = cu'$$

$$2. \frac{d}{dx}[u \pm v] = u' \pm v'$$

$$3. \frac{d}{dx}[uv] = u'v + uv'$$

$$4. \frac{d}{dx}\left[\frac{u}{v}\right] = \frac{u'v - uv'}{v^2}$$

$$5. \frac{d}{dx}[c] = 0$$

$$6. \frac{d}{dx}[u^n] = nu^{n-1}u'$$

$$7. \frac{d}{dx}[x] = 1$$

$$8. \frac{d}{dx}[|u|] = \frac{u}{|u|}(u'), u \neq 0$$

$$9. \frac{d}{dx}[\ln u] = \frac{u'}{u}$$

$$10. \frac{d}{dx}[e^u] = e^u u'$$

$$11. \frac{d}{dx}[\log_a u] = \frac{u'}{(\ln a)u}$$

$$12. \frac{d}{dx}[a^u] = (\ln a) a^u u'$$

$$13. \frac{d}{dx}[\sin u] = (\cos u)u'$$

$$14. \frac{d}{dx}[\cos u] = -(\sin u)u'$$

$$15. \frac{d}{dx}[\tan u] = (\sec^2 u)u'$$

$$16. \frac{d}{dx}[\cot u] = -(\csc^2 u)u'$$

$$17. \frac{d}{dx}[\sec u] = (\sec u \tan u)u'$$

$$18. \frac{d}{dx}[\csc u] = -(\csc u \cot u)u'$$

$$19. \frac{d}{dx}[\arcsin u] = \frac{u'}{\sqrt{1-u^2}}$$

$$20. \frac{d}{dx}[\arccos u] = \frac{-u'}{\sqrt{1-u^2}}$$

$$21. \frac{d}{dx}[\arctan u] = \frac{u'}{1+u^2}$$

$$22. \frac{d}{dx}[\operatorname{arc cot} u] = \frac{-u'}{1+u^2}$$

$$23. \frac{d}{dx}[\operatorname{arc sec} u] = \frac{u'}{|u|\sqrt{u^2-1}}$$

$$24. \frac{d}{dx}[\operatorname{arc csc} u] = \frac{-u'}{|u|\sqrt{u^2-1}}$$

$$\frac{d}{dx} x = \lim_{h \rightarrow 0} \frac{(x+h) - x}{h} = \lim_{h \rightarrow 0} \frac{h}{h} = 1$$

$$\frac{d}{dx} x^2 = \lim_{h \rightarrow 0} \frac{(x+h)^2 - x^2}{h} = \lim_{h \rightarrow 0} \frac{(\cancel{x^2} + 2x\cancel{h} + \cancel{h^2}) - \cancel{x^2}}{\cancel{h}} = 2x$$

$$\frac{d}{dx} x^3 = \lim_{h \rightarrow 0} \frac{(x+h)^3 - x^3}{h} = \lim_{h \rightarrow 0} \frac{(\cancel{x^3} + 3x^2\cancel{h} + 3x\cancel{h^2} + \cancel{h^3}) - \cancel{x^3}}{\cancel{h}} = 3x^2$$

We observe a pattern: 1 2x 3x² 4x³ 5x⁴ ...

Power Rule

$$\frac{d}{dx}(x^n) = nx^{n-1}$$

examples:

$$f(x) = x^4$$

$$f'(x) = 4x^3$$

$$\frac{d}{dx}(c) = 0$$

$$y = 3$$

If the derivative of a function is its slope, then for a constant function, the derivative must be zero.

$$y' = 0$$

Constant Multiple Rule

$$\frac{d}{dx}(cu) = c \frac{du}{dx}$$

$$\frac{d}{dx} 7x^5 = 7 \cdot 5x^4 = 35x^4$$

Sum Rule

$$\frac{d}{dx}(u + v) = \frac{du}{dx} + \frac{dv}{dx}$$

examples:

$$y = x^4 + 12x$$

$$y' = 4x^3 + 12$$

Difference Rule

$$\frac{d}{dx}(u - v) = \frac{du}{dx} - \frac{dv}{dx}$$

$$y = x^4 - 2x^2 + 2$$

$$y' = 4x^3 - 4x$$

Product Rule

$$\frac{d}{dx}(uv) = u \frac{dv}{dx} + v \frac{du}{dx}$$

or

$$d(uv) = u dv + v du$$

Notice that this is not just the product of two derivatives.

Examples:

$$\begin{aligned}\frac{d}{dx}[(x^2 + 3)(2x^3 + 5x)] &= (x^2 + 3)(6x^2 + 5) + (2x^3 + 5x)(2x) \\ &= 6x^4 + 5x^2 + 18x^2 + 15 + 4x^4 + 10x^2 \\ &= 10x^4 + 33x^2 + 15\end{aligned}$$

Quotient Rule

$$\frac{d}{dx} \left(\frac{u}{v} \right) = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

or

$$d \left(\frac{u}{v} \right) = \frac{v \, du - u \, dv}{v^2}$$

Examples:

$$\frac{d}{dx} \frac{2x^3 + 5x}{x^2 + 3} = \frac{(x^2 + 3)(6x^2 + 5) - (2x^3 + 5x)(2x)}{(x^2 + 3)^2}$$

Example:

Find the horizontal tangents of:

$$y = x^4 - 2x^2 + 2 \qquad \frac{dy}{dx} = 4x^3 - 4x$$

Horizontal tangents occur when slope = zero.

$$4x^3 - 4x = 0$$

$$x^3 - x = 0$$

$$x(x^2 - 1) = 0$$

$$x(x+1)(x-1) = 0$$

$$x = 0, -1, 1$$

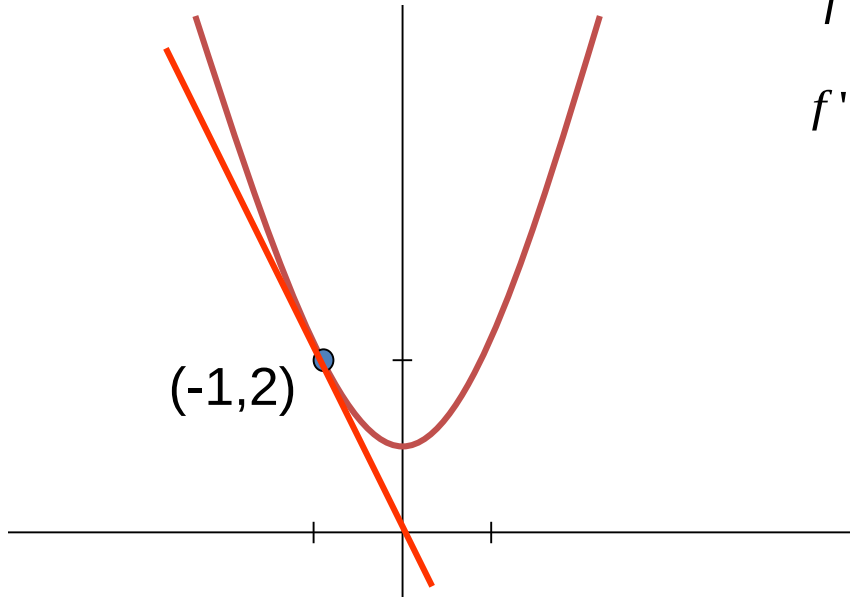
Plugging the x values into the original equation, we get:

$$y = 2, \quad y = 1, \quad y = 1$$

(The function is even, so we only get two horizontal tangents.)

Example:

Find the slope of the graph of $f(x) = x^2 + 1$ at the point $(-1, 2)$.
Then, find the equation of the tangent line.



$$f(x) = x^2 + 1$$

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

$$= \lim_{\Delta x \rightarrow 0} \frac{(x + \Delta x)^2 + 1 - (x^2 + 1)}{\Delta x}$$

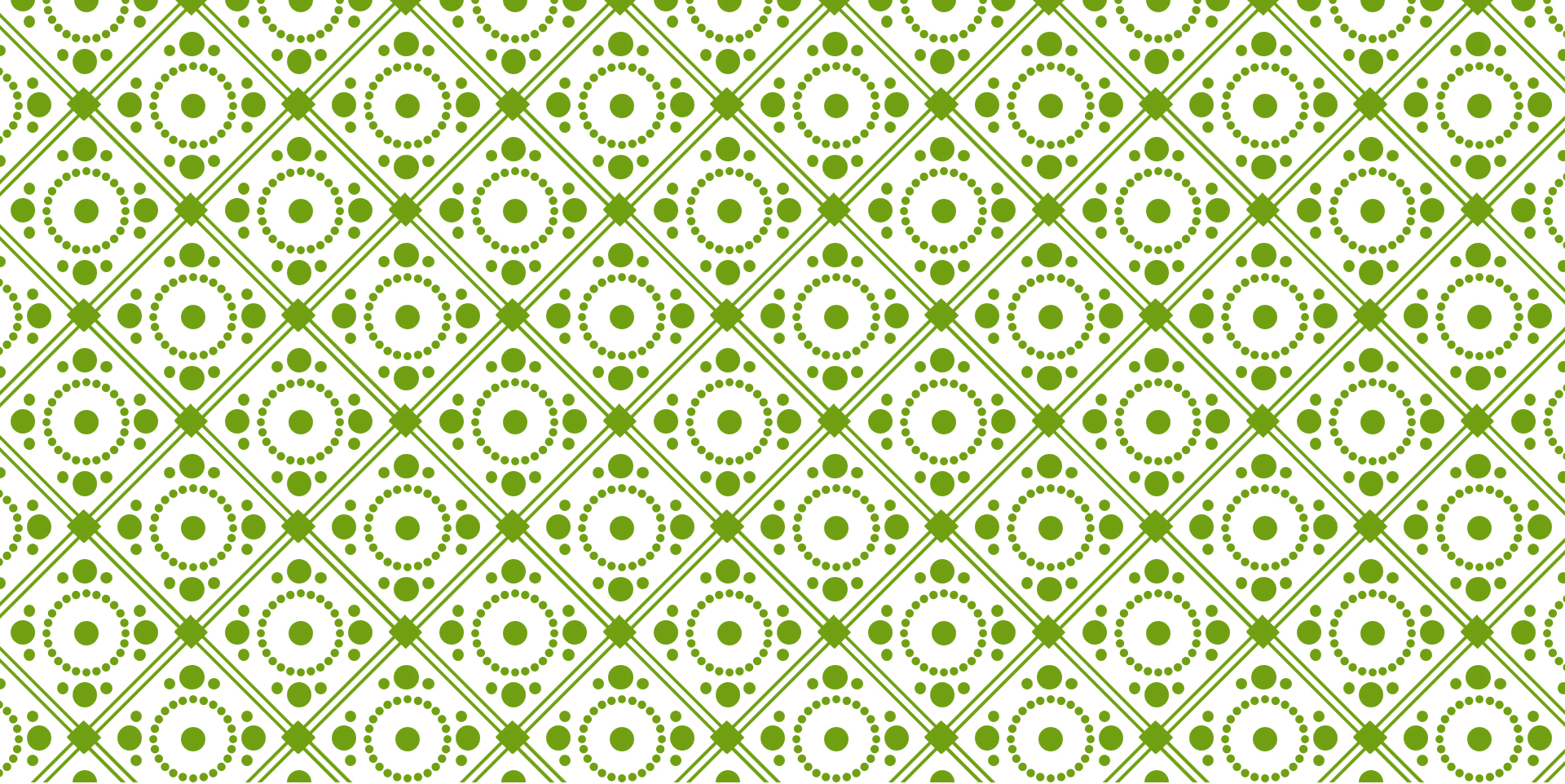
$$= \lim_{\Delta x \rightarrow 0} \frac{\cancel{x^2} + 2x\Delta x + (\Delta x)^2 + \cancel{1} - \cancel{x^2} - \cancel{1}}{\Delta x}$$

$$= \lim_{\Delta x \rightarrow 0} \frac{\cancel{\Delta x}(2x + \Delta x)}{\cancel{\Delta x}} = 2x$$

Therefore, the slope at any point $(x, f(x))$ is given by $m = 2x$

What is the slope at the point $(-1, 2)$? $m = 2 \cdot (-1) = -2$

The equation of the tangent line is $y - 2 = -2(x + 1)$



DERIVATIVES

HIGH ORDER DERIVATIVES

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2021 - 2022

High Order Derivatives

$y' = \frac{dy}{dx}$ is the first derivative of y with respect to x .

$y'' = \frac{dy'}{dx} = \frac{d}{dx} \frac{dy}{dx} = \frac{d^2 y}{dx^2}$ is the second derivative.
(y double prime)

$y''' = \frac{dy''}{dx}$ is the third derivative.

$y^{(4)} = \frac{d}{dx} y'''$ is the fourth derivative.

We will learn later
what these higher
order derivatives
are used for.

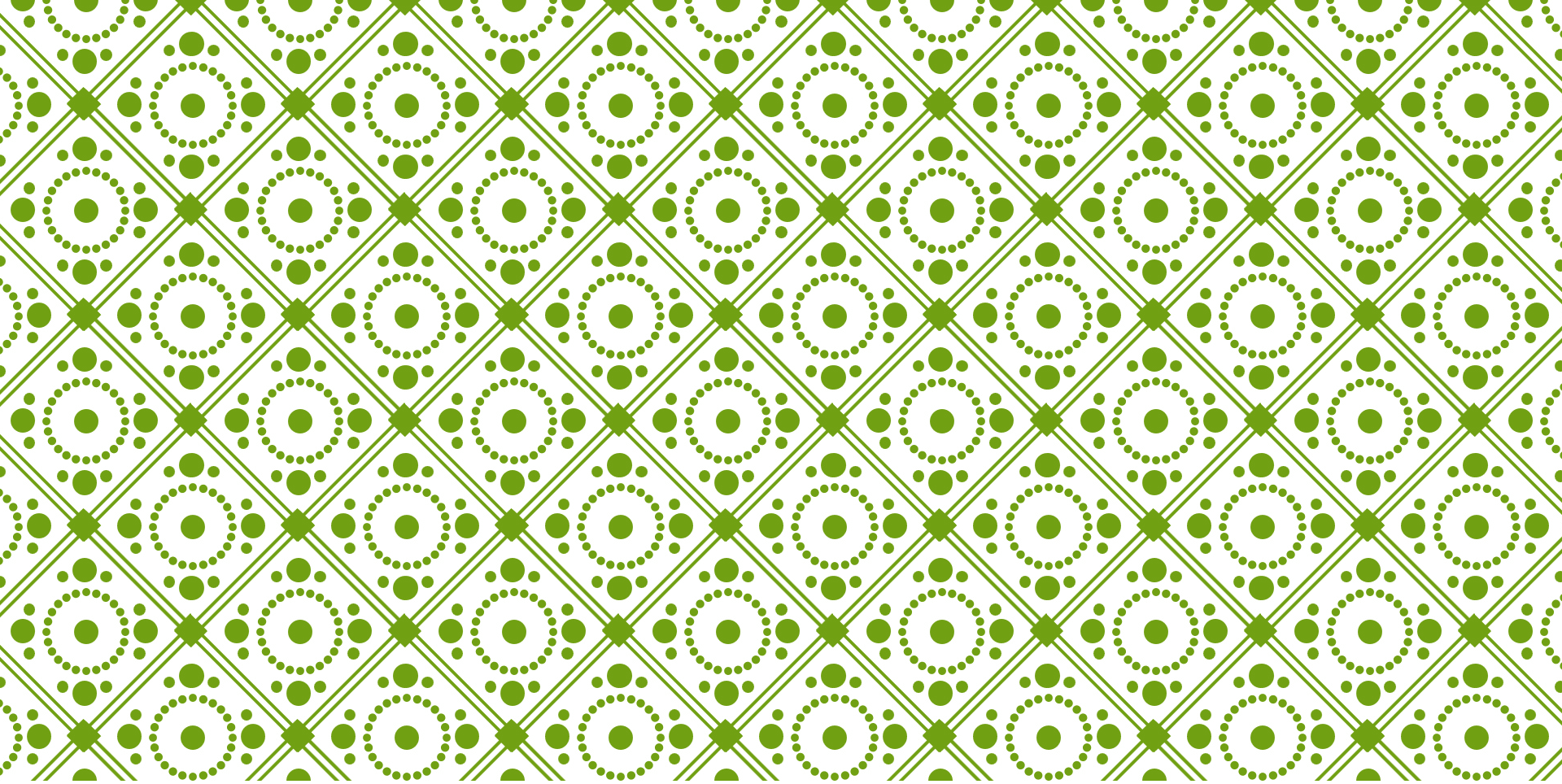
Example:

$$f(x) = \frac{x}{x-1}$$

$$f'(x) = \frac{(x-1)(1) - x(1)}{(x-1)^2} = \frac{-1}{(x-1)^2}$$

$$f''(x) = \frac{d}{dx} \left(\frac{-1}{(x-1)^2} \right) = \frac{d}{dx} \left((x-1)^{-2} \right)$$

$$f''(x) = -2(x-1)^{-3}(1) = \frac{-2}{(x-1)^3}$$



DERIVATIVES

TECHNIQUES OF DIFFERENTIATION

Nyimas Dewi Sartika
Universitas Multimedia Nusantara
2021 - 2022

Composition of Functions (Review)

Definition:

If $f(x)$ and $g(x)$ are both differentiable and $F(x)$ is the composite function defined by $F(x) = f \circ g(x) = f(g(x))$

Example: If $f(x) = \tan x$ and $g(x) = 3x+2$, then $f(g(x)) = f(3x+2) = \tan(3x+2)$

Each function below represents $F(x) = f(g(x))$, define $f(x)$ and $g(x)$

No	$F(x)$	$f(x)$	$g(x)$
1	$y = 3\sqrt{x^3 - 4x}$	$3\sqrt{x}$	$x^3 - 4x$
2	$y = \frac{1}{(2x - 3)^3}$	$\frac{1}{x^3}$	$2x - 3$
3	$y = \sin(3x)$	$\sin x$	$3x$
4	$y = (3x - 8)^4$	x^4	$3x - 8$
5	$y = \csc^2(2x - \pi)$	$\csc^2 x$	$2x - \pi$

Chain Rule

If $f(x)$ and $g(x)$ are both differentiable and $F(x)$ is the composite function defined by $F(x) = f(g(x))$, then $F(x)$ is differentiable and $F'(x)$ is given by:

$$\frac{d}{dx}[f(g(x))] = f'(g(x)) \cdot g'(x)$$

In Leibniz notation, if $y = f(u)$ and $u = g(x)$ are both differentiable functions, then

$$\frac{df}{dx} = \frac{df}{du} \cdot \frac{du}{dx}$$

Guidelines for Chain Rule

For $f(u(x))$ to find $f'(x)$:

- Step 1: Decompose the function: $f(u)$ and $u(x)$
- Step 2: Differentiate the mother function: $f'(u)$
- Step 3: Differentiate the composed function: $u'(x)$
- Step 4: Multiply the resultant derivatives:

$$f'(x) = f'(u) \cdot u'(x)$$

- Step 5: Substitute for u and Simplify

Examples:

1. $y = (3x - 5x^2)^7$

2. $y = \frac{1}{(2x - 3)^3}$

3. $y = \frac{3}{\sqrt{x^3 + 4x}}$

4. $y = x^2 \sqrt{x - 3}$

5. $y = \cos^3(\sin \sqrt{2x-1})$

6. $y = (2x-1)^3(x^2+3)^4$

7. $y = \frac{(2x-1)^3}{(x^2+3)^4}$

8. $f(t) = \left(\frac{2t-1}{t^2+3} \right)^5$

Implicit Differentiation

The functions we considered so far can be described by expressing one variable explicitly in terms of another variable, e.g.

$$y = x^2 + 3x \quad \text{or} \quad y = \cos x$$

However, some functions are defined implicitly by a relation between x and y such as

$$x^2 + y^2 = 1 \quad \text{or} \quad 2y = x^2 + \sin y$$

For most of this kind of functions, it's not easy to solve for y to express it explicitly as a function of x .

But fortunately we don't need to solve an equation for y in order to find the derivative of y . Instead we can use the method of implicit differentiation:

Guidelines for Implicit Differentiation

Example 1: Find $\frac{dy}{dx}$ if $x^2 + y^2 = 1$

- Step 1: Differentiate both sides of the equation with respect to x , treating y as a function of x . This requires the chain rule.

$$\frac{d}{dx} x^2 + \frac{d}{dx} y^2 = \frac{d}{dx} 1$$

- Step 2: Collect terms with dy/dx on one side of the equation.

$$2x + 2y \frac{dy}{dx} = 0$$

- Step 3: Factor dy/dx

$$2y \frac{dy}{dx} = -2x$$

- Step 4: Solve for dy/dx

$$\frac{dy}{dx} = -\frac{x}{y}$$

Example 2: Find $\frac{dy}{dx}$ if $y^2 = x^2 + \sin(xy)$

1. Differentiate both sides of the equation with respect to x , treating y as a function of x . This requires the chain rule.

$$2y \frac{dy}{dx} = 2x + \cos(xy) \left(x \frac{dy}{dx} \right) + \cos(xy) y$$

2. Collect terms with dy/dx on one side of the equation.

$$2y \frac{dy}{dx} - \cos(xy) \left(x \frac{dy}{dx} \right) = 2x + \cos(xy) y$$

3. Factor dy/dx

$$\frac{dy}{dx} (2y - x \cos(xy)) = 2x + y \cos(xy)$$

4. Solve for dy/dx

$$\frac{dy}{dx} = \frac{2x + y \cos(xy)}{2y - x \cos(xy)}$$

Example 3: Find $\frac{d^2y}{dx^2}$ if $2x^3 - 3y^2 = 7$

Example 4: Find the equation of the tangent line to the curve at $(-1,2)$

$$x^2 - xy + y^2 = 7$$

Example 5: Find $\frac{d^2y}{dx^2}$ if $x^2 - xy + y^2 = 7$

Example 6: Find $\frac{d^2y}{dx^2}$ if $x + y = (x^2 + y^2)^2$

Derivative of Parametric Equation

Consider parametric equations: $x = f(t)$ and $y = g(t)$

First derivative

$$\frac{dy}{dx} = \frac{dy / dt}{dx / dt}$$

Second derivative

$$\frac{d^2 y}{dx^2} = \frac{\frac{d}{dt} \left(\frac{dy}{dx} \right)}{\frac{dx}{dt}}$$

Example 1: Find the second derivative of the parametric equations $x = 3 + 4 \cos t$ and $y = 1 - \sin t$

- First derivative: $\frac{dy}{dx} = \frac{-\cos t}{-4 \sin t} = \frac{1}{4} \cot t$
- Second derivative: $\frac{d^2 y}{dx^2} = \frac{1}{4} \cdot \frac{-\csc^2 t}{-4 \sin t} = \frac{1}{16 \sin^3 t}$

Example 2:

A curve is defined by the parametric equations

$$x = t^2 + \frac{1}{t^2}, \quad y = t^2 - \frac{1}{t^2} \quad (t \neq 0)$$

Find an expression for $\frac{dy}{dx}$ in terms of t , simplify your answer

$$\begin{aligned} x &= t^2 + \frac{1}{t^2} & \frac{dx}{dt} &= 2t - 2t^{-3} \\ y &= t^2 - \frac{1}{t^2} & \frac{dy}{dt} &= 2t + 2t^{-3} \end{aligned} \qquad \frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{2t + 2t^{-3}}{2t - 2t^{-3}} = \frac{t^4 + 1}{t^4 - 1}$$

Example 3:

A curve is defined by the parametric equations

$$x = \frac{t}{1+t}, \quad y = \frac{1+t}{1-t} \quad (t \neq 0)$$

Find an expression for $\frac{dy}{dx}$ in terms of t , simplify your answer

$$x = \frac{t}{1+t} \quad \frac{dx}{dt} = \frac{(1+t) - t(1)}{(1+t)^2} = \frac{1}{(1+t)^2}$$

$$y = \frac{1+t}{1-t} \quad \frac{dy}{dt} = \frac{(1)(1-t) - (1+t)(-1)}{(1-t)^2} = \frac{1-t+1+t}{(1-t)^2} = \frac{2}{(1-t)^2}$$

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{\frac{2}{(1-t)^2}}{\frac{1}{(1+t)^2}} = \frac{2}{(1-t)^2} \times \frac{(1+t)^2}{1} = 2 \frac{(1+t)^2}{(1-t)^2} = 2 \left(\frac{1+t}{1-t} \right)^2$$

Example 4:

$$x = t^2 \quad y = 2t$$

$$\frac{dx}{dt} = 2t \quad \frac{dy}{dt} = 2$$

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{2}{2t} = \frac{1}{t}$$

$$\frac{d^2y}{dx^2} = \frac{d}{dt} \left(\frac{1}{t} \right) \frac{dt}{dx}$$

$$\frac{d}{dt} (t^{-1}) = -t^{-2} = -\frac{1}{t^2} \quad \frac{dt}{dx} = \frac{1}{\frac{dx}{dt}} = \frac{1}{2t}$$

$$\frac{d^2y}{dx^2} = -\frac{1}{t^2} \times \frac{1}{2t} = -\frac{1}{2t^3}$$

Inverse Functions (Review)

Definition:

Let f be a bijective function with domain A and range B . Then the inverse function f^{-1} has domain B and range A and is defined by

$$f^{-1}(y) = x \iff f(x) = y \quad \text{for any } y \text{ in } B.$$

Note: $f^{-1}(x)$ does not mean $1/f(x)$

$$f^{-1}(f(x)) = x \quad \text{for every } x \text{ in } A$$

$$f(f^{-1}(x)) = x \quad \text{for every } x \text{ in } B$$

Guidelines for Inverse Function

- Step 1: Write $y = f(x)$
- Step 2: Solve this equation for x in terms of y (if possible)
- Step 3: To express f^{-1} as a function of x , interchange x and y
- Step 4: The resulting equation is $y = f^{-1}(x)$

Example:

Find the inverse of $f(x) = 5 - x^3$

$$y = 5 - x^3$$

$$x^3 = 5 - y$$

$$x = \sqrt[3]{5 - y}$$

$$f^{-1}(y) = \sqrt[3]{5 - y}$$

$$f^{-1}(x) = \sqrt[3]{5 - x}$$

Example:

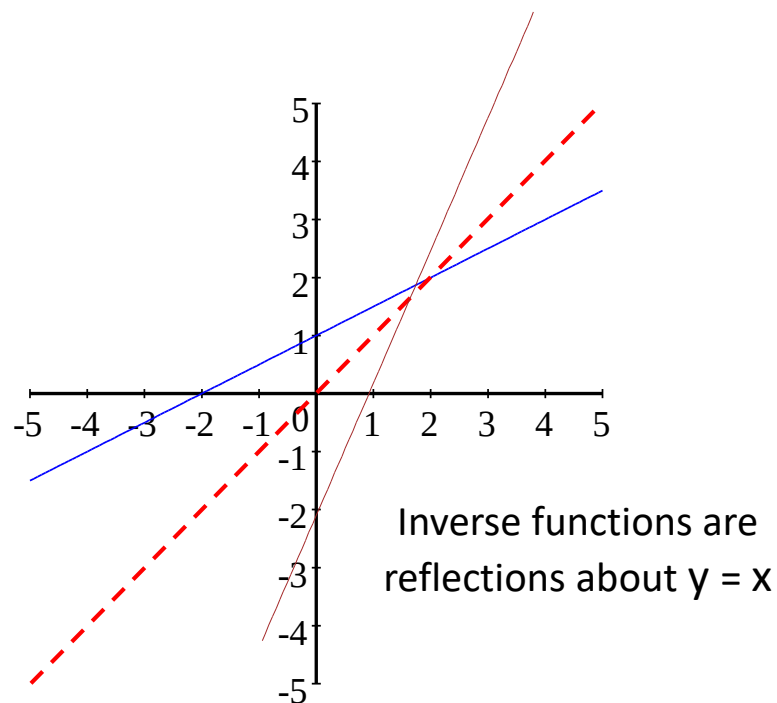
$$f(x) = \frac{1}{2}x + 1$$

$$y = \frac{1}{2}x + 1$$

$$x = 2y - 2$$

$$f^{-1}(y) = 2y - 2$$

$$f^{-1}(x) = 2x - 2$$

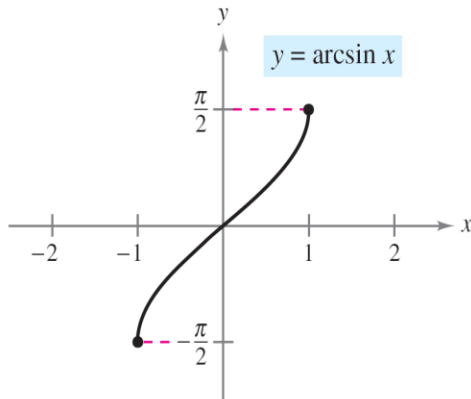


Under suitable restrictions, each of the six trigonometric functions is one-to-one and so has an inverse function, as shown in the following definition.

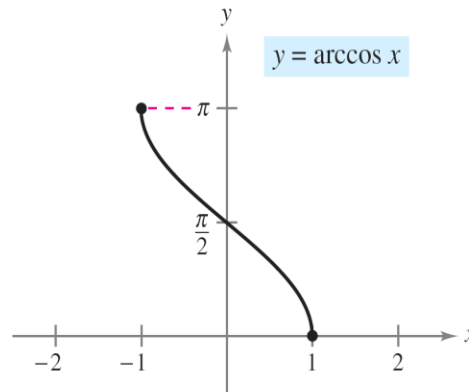
DEFINITIONS OF INVERSE TRIGONOMETRIC FUNCTIONS

<u>Function</u>	<u>Domain</u>	<u>Range</u>
$y = \arcsin x \text{ iff } \sin y = x$	$-1 \leq x \leq 1$	$-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$
$y = \arccos x \text{ iff } \cos y = x$	$-1 \leq x \leq 1$	$0 \leq y \leq \pi$
$y = \arctan x \text{ iff } \tan y = x$	$-\infty < x < \infty$	$-\frac{\pi}{2} < y < \frac{\pi}{2}$
$y = \operatorname{arccot} x \text{ iff } \cot y = x$	$-\infty < x < \infty$	$0 < y < \pi$
$y = \operatorname{arcsec} x \text{ iff } \sec y = x$	$ x \geq 1$	$0 \leq y \leq \pi, \quad y \neq \frac{\pi}{2}$
$y = \operatorname{arccsc} x \text{ iff } \csc y = x$	$ x \geq 1$	$-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}, \quad y \neq 0$

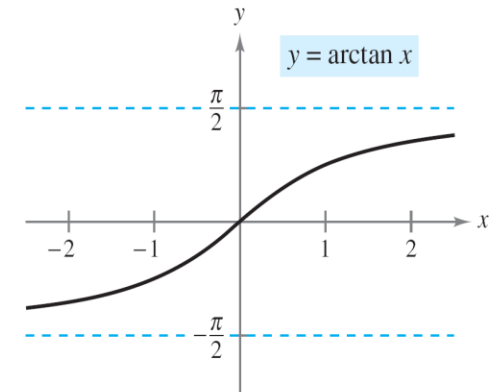
The graphs of the six inverse trigonometric functions



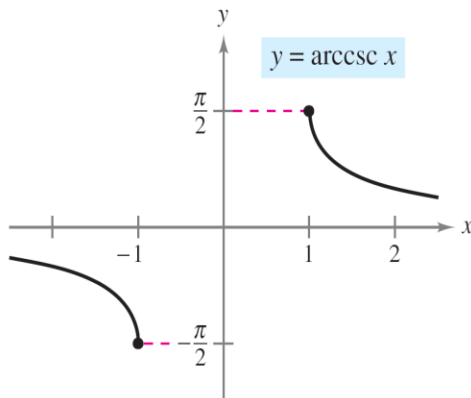
Domain: $[-1, 1]$
Range: $[-\pi/2, \pi/2]$



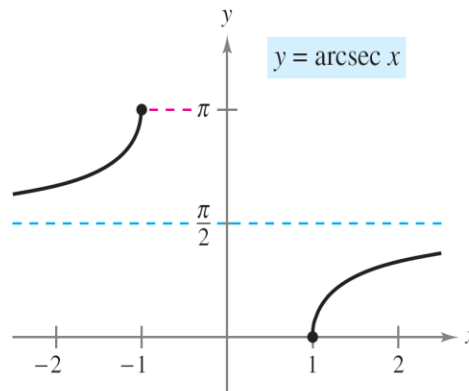
Domain: $[-1, 1]$
Range: $[0, \pi]$



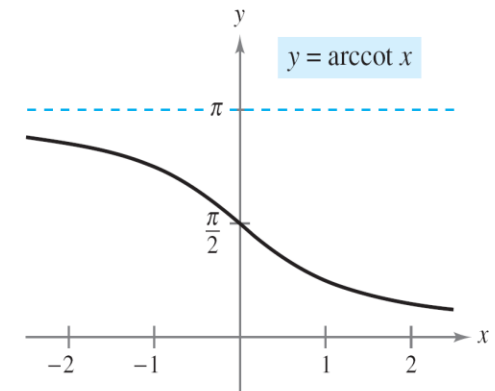
Domain: $(-\infty, \infty)$
Range: $(-\pi/2, \pi/2)$



Domain: $(-\infty, -1] \cup [1, \infty)$
Range: $[-\pi/2, 0) \cup (0, \pi/2]$



Domain: $(-\infty, -1] \cup [1, \infty)$
Range: $[0, \pi/2) \cup (\pi/2, \pi]$



Domain: $(-\infty, \infty)$
Range: $(0, \pi)$

PROPERTIES OF INVERSE TRIGONOMETRIC FUNCTIONS

If $-1 \leq x \leq 1$ and $-\pi/2 \leq y \leq \pi/2$, then

$$\sin(\arcsin x) = x \quad \text{and} \quad \arcsin(\sin y) = y.$$

If $-\pi/2 < y < \pi/2$, then

$$\tan(\arctan x) = x \quad \text{and} \quad \arctan(\tan y) = y.$$

If $|x| \geq 1$ and $0 \leq y < \pi/2$ or $\pi/2 < y \leq \pi$, then

$$\sec(\operatorname{arcsec} x) = x \quad \text{and} \quad \operatorname{arcsec}(\sec y) = y.$$

Similar properties hold for the other inverse trigonometric functions.

Inverse functions have the properties: $f(f^{-1}(x)) = x$ and $f^{-1}(f(x)) = x$.

When applying these properties to inverse trigonometric functions, remember that the trigonometric functions have inverse functions only in restricted domains.

For x -values outside these domains, these two properties do not hold.

For example, $\arcsin(\sin \pi)$ is equal to 0, not π .

Examples:

1. **a.** $\arcsin\left(-\frac{1}{2}\right)$ **b.** $\arccos 0$ **c.** $\arctan \sqrt{3}$ **d.** $\arcsin(0.3)$

a. By definition, $y = \arcsin\left(-\frac{1}{2}\right)$ implies that $\sin y = -\frac{1}{2}$.

In the interval $[-\pi/2, \pi/2]$, the correct value of y is $-\pi/6$.

2. $\arctan(2x - 3) = \frac{\pi}{4}$

$$\tan(\arctan(2x - 3)) = \tan \frac{\pi}{4}$$

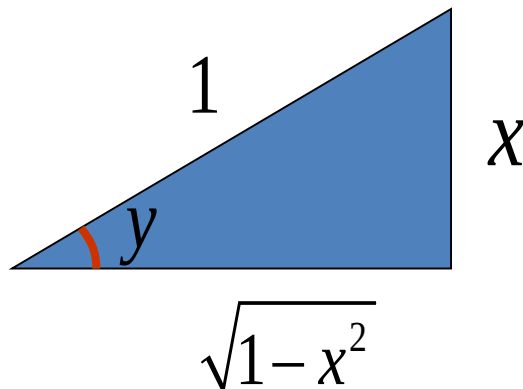
$$(2x - 3) = 1$$

$$x = 2$$

3. Given $y = \arcsin x$, where $0 < y < \pi / 2$, find $\cos y$.

$$\sin y = x$$

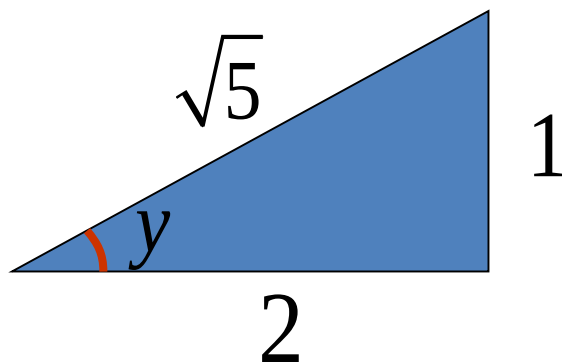
$$\cos y = \sqrt{1 - x^2}$$



4. Given $y = \operatorname{arcsec}(\sqrt{5}/2)$, find $\tan y$.

$$\sec y = \frac{\sqrt{5}}{2}$$

$$\tan y = \frac{1}{2}$$



Derivatives of Inverse Trig Functions

Let u be a differentiable function of x .

$$\frac{d}{dx} [\arcsin u] = \frac{u'}{\sqrt{1-u^2}}$$

$$\frac{d}{dx} [\arccos u] = \frac{-u'}{\sqrt{1-u^2}}$$

$$\frac{d}{dx} [\arctan u] = \frac{u'}{1+u^2}$$

$$\frac{d}{dx} [\operatorname{arccot} u] = \frac{-u'}{1+u^2}$$

$$\frac{d}{dx} [\operatorname{arcsec} u] = \frac{u'}{|u|\sqrt{u^2-1}}$$

$$\frac{d}{dx} [\operatorname{arccsc} u] = \frac{-u'}{|u|\sqrt{u^2-1}}$$

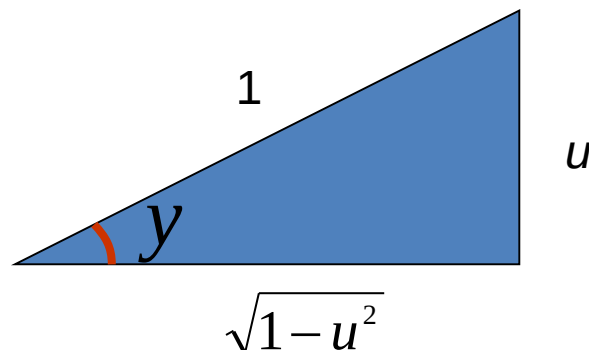
Examples:

5. $y = \sin^{-1} u$, Find y'

$$\sin y = u$$

$$(\cos y) y' = u'$$

$$y' = \frac{u'}{(\cos y)} \qquad y' = \frac{u'}{\sqrt{1-u^2}}$$



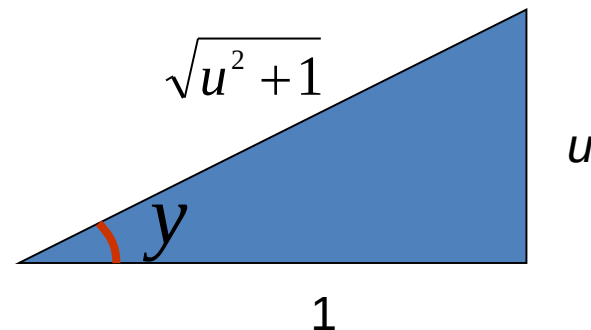
6. $y = \tan^{-1} u$, Find y'

$$\tan y = u$$

$$(\sec^2 y) y' = u'$$

$$y' = \frac{u'}{\sec^2 y} = \frac{u'}{(\sqrt{u^2+1})^2}$$

$$y' = \frac{u'}{1+u^2}$$



$$y = \sin^{-1} x$$

$$\sin y = x$$

$$\frac{d}{dx} \sin y = \frac{d}{dx} x$$

$$\cos y \frac{dy}{dx} = 1$$

$$\frac{dy}{dx} = \frac{1}{\cos y}$$

$$\frac{dy}{dx} = \frac{1}{\sqrt{1 - \sin^2 y}}$$

$$\sin^2 y + \cos^2 y = 1$$

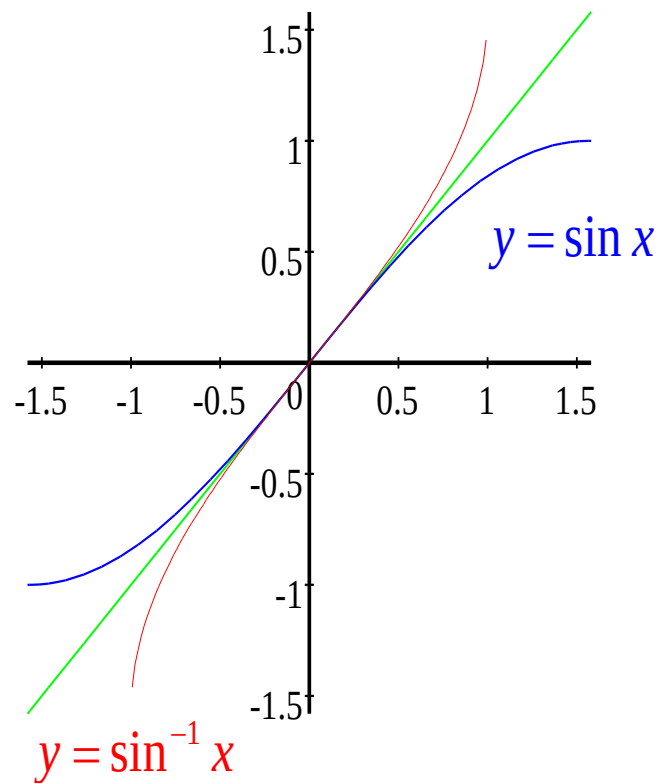
$$\cos^2 y = 1 - \sin^2 y$$

$$\cos y = \pm \sqrt{1 - \sin^2 y}$$

But $-\frac{\pi}{2} < y < \frac{\pi}{2}$

So $\cos y$ is positive.

$$\therefore \cos y = \sqrt{1 - \sin^2 y}$$



$$\begin{cases} \text{Domain: } -1 \leq x \leq 1 \\ \text{Range: } -\pi/2 \leq y \leq \pi/2 \end{cases}$$

Examples:

1. $y = \arcsin(5x + 2)$

2. $y = \arctan(x^2 - 5)$

3. $y = \csc^{-1}(x^3 - 2x + 1)$

4. $y = \cot^{-1}\left(\frac{x}{6}\right)$

5. $y = 8 \arcsin x + 7 \arccos x$

6. $y = 4 \arccos(3x + 9)$

7. $y = -4 \tan^{-1}\left(\frac{x}{8}\right)$

8. Find the equation of the tangent line to the graph of the function at the given point:

$$y = \arctan\left(\frac{x}{2}\right) \quad \text{at} \quad \left(2, \frac{\pi}{4}\right)$$

Exponential and Logarithmic Functions (Review)

Consider $f(x) = a^x$ where $a > 0$ and $a \neq 1$.

This is a one-to-one function, therefore it has an inverse.

The inverse is called the logarithmic function with base a .

$$y = a^x \Leftrightarrow x = \log_a y$$

The most commonly used bases for logs are 10 and e

- base 10: $\log_{10} x = \log x$
- base e: $\log_e x = \ln x$

Examples: $16 = 2^4 \Leftrightarrow 4 = \log_2 16$

RULES:

1. $a^{\log_a x} = x \iff \log_a a^x = x \quad (a > 0, a \neq 1, x > 0)$
(log and exp are inverse functions)
2. Product rule: $\log_a xy = \log_a x + \log_a y$
3. Quotient rule: $\log_a \frac{x}{y} = \log_a x - \log_a y$
4. Power rule: $\log_a x^y = y \log_a x$
5. Change of base formula: $\log_a x = \frac{\ln x}{\ln a}$

<i>Original Rule</i>	<i>Generalized Rule</i>	<i>In Words</i>
$\frac{d}{dx} [e^x] = e^x$	$\frac{d}{dx} [e^u] = e^u \frac{du}{dx}$	<i>The derivative of e raised to a quantity is e raised to that quantity, times the derivative of that quantity.</i>
$\frac{d}{dx} [b^x] = b^x \ln b$	$\frac{d}{dx} [b^u] = b^u \ln b \frac{du}{dx}$	<i>The derivative of b raised to a quantity is b raised to that quantity, times $\ln b$, times the derivative of that quantity.</i>

Examples:

$$1. \quad \frac{d}{dx} [3e^x] = 3 \frac{d}{dx} [e^x] = 3e^x$$

$$2. \quad \frac{d}{dx} [3^x] = 3^x \ln 3$$

$$\begin{aligned}
 3. \quad \frac{d}{dx} [e^{x^2+1}] &= e^{x^2+1} \frac{d}{dx} [x^2 + 1] \\
 &= e^{x^2+1} (2x) = 2x e^{x^2+1}
 \end{aligned}$$

<i>Original Rule</i>	<i>Generalized Rule</i>	<i>In Words</i>
$\frac{d}{dx}[\ln x] = \frac{1}{x}$	$\frac{d}{dx}[\ln u] = \frac{1}{u} \frac{du}{dx}$	<i>The derivative of the natural logarithm of a quantity is 1 over that quantity, times the derivative of that quantity.</i>
$\frac{d}{dx}[\log_b x] = \frac{1}{x \ln b}$	$\frac{d}{dx}[\log_b u] = \frac{1}{u \ln b} \frac{du}{dx}$	<i>The derivative of the log to base b of a quantity is 1 over the product of ln b and that quantity, times the derivative of that quantity.</i>

Examples:

- $\frac{d}{dx}[3 \ln x] = 3 \cdot \frac{1}{x} = \frac{3}{x}$
- $\frac{d}{dx}[\log_3 x] = \frac{1}{x \ln 3} \approx \frac{1}{1.0986x}$
- $$\begin{aligned} \frac{d}{dx} \ln[x^2 + 1] &= \frac{1}{x^2 + 1} \frac{d}{dx}(x^2 + 1) \\ &= \frac{1}{x^2 + 1} (2x) = \frac{2x}{x^2 + 1} \end{aligned}$$

Examples: Prove that $\frac{d}{dx}(\ln x) = \frac{1}{x}$

$$\begin{aligned}\frac{d}{dx}[\ln x] &= \lim_{h \rightarrow 0} \frac{\ln(x+h) - \ln x}{h} \\&= \lim_{h \rightarrow 0} \frac{1}{h} [\ln(x+h) - \ln x] \\&= \lim_{h \rightarrow 0} \frac{1}{h} \ln \left(\frac{x+h}{x} \right) \\&= \lim_{h \rightarrow 0} \frac{1}{h} \ln \left(1 + \frac{h}{x} \right) \\&= \lim_{h \rightarrow 0} \ln \left(1 + \frac{h}{x} \right)^{1/h} \\&= \lim_{h \rightarrow 0} \ln \left[\left(1 + \frac{1}{(x/h)} \right)^{x/h} \right]^{1/x} . \\&\ln[e]^{1/x} = \frac{1}{x} \ln e = \frac{1}{x}\end{aligned}$$

As $h \rightarrow 0^+$, the quantity x/h is getting large and positive,
and so the quantity in brackets is approaching e

Logarithmic Differentiation

- Given $y = \sqrt[18]{(x^{10} + 1)^3 (x^7 - 3)^8}$
- Take the log of both sides, simplify
- Now differentiate both sides with respect to x , solve for dy/dx

Guidelines for Logarithmic Differentiation

- Step 1: Take natural logarithms of both sides of an equation $y = f(x)$ and use the properties of logarithms to simplify.
- Step 2: Differentiate implicitly with respect to x
- Step 3: Solve the resulting equation for y'

The calculation of derivatives of complicated functions involving products, quotients, or powers can often be simplified by taking logarithms.

Examples:

1. $y = \sqrt[18]{(x^{10} + 1)^3 (x^7 - 3)^8}$